

Curriculum Vitae for Ghulam Rasool, PhD

Date: February 27, 2026

Current Position

Associate Member

Jan 2026–Present

Department of Machine Learning

Department of Neuro-Oncology

Department of Diagnostic Imaging and Interventional Radiology

Cancer Epidemiology Program

Moffitt Cancer Center & Research Institute

12902 Magnolia Drive, Tampa, FL 33612

☎ (813) 745-3161 ✉ ghulam.rasool@moffitt.org 🔗 [GitHub](#) [Hugging Face](#) 🎓 [Google Scholar](#)

Current Academic Appointments

Assistant Professor

Department of Oncologic Sciences, Morsani College of Medicine

University of South Florida, Tampa, FL

Assistant Professor (Courtesy)

Department of Electrical Engineering

University of South Florida, Tampa, FL

Education

- **Ph.D., Systems Engineering** 2011–2014
University of Arkansas at Little Rock (UALR), Little Rock, AR
- **M.Sc., Computer Engineering** 2008–2010
Centre for Advanced Studies in Engineering (CASE), Islamabad, Pakistan
- **B.E., Mechanical Engineering** 1996–1999
National University of Sciences and Technology (NUST), Islamabad, Pakistan

Postgraduate Training and Fellowship Appointments

- **Postdoctoral Fellow** 2014–2016
Shirley Ryan AbilityLab (formerly Rehabilitation Institute of Chicago), Chicago, IL
- **Postdoctoral Fellow (Joint Appointment)** 2014–2016
Department of Physical Medicine & Rehabilitation, Feinberg School of Medicine, Northwestern University, Chicago, IL
- **Research Scholar** 2017–2018
Department of Electrical and Computer Engineering, Rowan University, Glassboro, NJ

Academic Appointments and Employment

- **Assistant Member (Tenure Track)** 2022–2025
Department of Machine Learning, Moffitt Cancer Center, Tampa, FL
- **Assistant Professor (Tenure Track)** 2019–2022
Department of Electrical and Computer Engineering, Rowan University, Glassboro, NJ
- **Lecturer / Adjunct Lecturer** 2017–2019

Certifications

- **2024** — NVIDIA Efficient Large Language Model (LLM) Customization
- **2024** — NVIDIA Building LLM Applications with Prompt Engineering
- **2024** — NVIDIA Rapid Application Development with LLMs
- **2022** — NVIDIA Building Transformer-based Natural Language Processing
- **2018** — NVIDIA Fundamentals of Deep Learning

Honors and Awards

- **2024** — Junior Researcher of the Year, Quantitative Science, Moffitt Cancer Center
- **2018** — NSF I-Corps Team Award, Rowan University
- **2018** — Google Faculty Institute Travel Award, Google Inc.
- **2014** — Brinson Foundation Postdoctoral Fellowship, Rehabilitation Institute of Chicago
- **2014** — Student Research Expo, Best Student Poster (3rd Position), UALR
- **2013** — Outstanding Doctoral Student, UALR
- **2013** — Student Research Expo, Best Student Poster (1st Position), UALR
- **2012** — Outstanding Doctoral Student, UALR
- **2012** — Student Research Expo, Best Student Poster (2nd Position), UALR
- **2012** — National Science Foundation Travel Grant, IEEE GENSIPS, UALR
- **2011** — Graduate Assistantship Award, UALR

Media and Press Highlights

- **2026** — “Using AI for pathology report data extraction”, Marilyn Bui and **Ghulam Rasool**, *Daily Path News*, [Poscast Link](#)
- **2025** — “As Cancer Patients Use AI Chatbots, Dangers Lurk—but New Safeguards Can Help”, **Ghulam Rasool**, Brian D. Gonzalez, and Katherine Chung-Bridges, *The Cancer Letter*, Vol. 51, No. 25, [Publication Link](#)
- **2025** — Pathology News. [Beyond the Black Box: New Study Reveals How AI Models Actually ‘Think’ Through Pathology Cases.](#)
- **2025** — AI in Pathology: The Six Pillars of Progress. [Generative tools are adding synthetic data, seamless reporting, and multimodal analysis to an already expanding AI toolkit.](#)
- **2025** — Multimodal AI for Cachexia Detection, AACR Annual Meeting 2025 news release: [Improving Cancer Care With Artificial Intelligence](#), [Daily Takeaways From the AACR Annual Meeting 2025: Sunday, April 27](#)
- **2025** — Multimodal AI for Cachexia Detection, featured in Moffitt Endeavor Newsletter: [New AI Tool Offers Hope for Early Detection of Cancer Cachexia](#)
- **2025** — Safety and Reliability of Chatbots, featured in Moffitt Endeavor Newsletter: [5 Things To Know Before Asking a Chatbot About Cancer](#)
- **2024** — Improving Radiology Report Conciseness and Structure via Local Large Language Model, highlighted by Radiological Society of North America (RSNA): [Demystifying Radiology Reports with ChatGPT](#)

Research Support

Ongoing Research Support (Total Count: 13)

- 1. AI-Driven Early Detection of Cachexia in Pancreatic Cancer and Feasibility of Diet and Exercise Interventions**
Florida Department of Health, Florida Cancer Innovation Fund
Rasool (Contact PI), Crowder, Permuth, Parker (MPIs)
06/2025–05/2026 | \$1.06M | 25% Effort
- 2. Advancing Health Research Through Ethical, Multimodal AI - MedGemma Federated Clinical Trial Matching**
University of North Carolina at Chapel Hill/ Prime: NIH/ODSS
Rasool (Site PI)
09/2025–08/2026 | \$183,625 | 10% Effort
- 3. Multimodal Federated Learning for Pan-Cancer Detection of Cancer Cachexia and Improved Risk Stratification of IPMNs**
NIH P30 CA076292 (CCSG Administrative Supplement)
Rasool (Project PI), Cleveland (Overall PI)
10/2024–01/2026 | \$202K | 7.5% Effort
- 4. Trustworthy Artificial Intelligence for the Volumetric Evaluation of Brain Tumors**
NSF PFI #2234468
Rasool (PI)
08/2023–07/2025 | \$250K | 15% Effort
- 5. Automated De-Identification of Radiology and Pathology Data**
NIH/NCI SBIR Phase II
Rasool (Sub-award PI), IBIS Inc. (Prime)
08/2023–07/2025 | \$1.998M (Moffitt: \$521K) | 20% Effort
- 6. Quantitative Imaging Clinical Validation Center at Moffitt Cancer Center**
NIH U01 CA200464
Rasool (Co-Investigator), Heini, Schabath (MPIs)
09/2023–08/2028 | \$6.218M | 5% Effort
- 7. Non-invasive Radiomic Biomarkers to Predict Treatment Response for Immunotherapy of Lung Cancer**
Florida Biomedical Research Program (FBRP), Bankhead-Coley
Rasool (Co-Investigator), Schabath (PI)
06/2021–04/2026 | \$1.327M | 5% Effort
- 8. Applying Large Language Models to Accelerate Abstraction of Cancer Pathology Reports for Cancer Registry**
NIH P30 CA076292-25S4 (Administrative Supplement)
Rasool (Co-Investigator), Cleveland (PI), Thieu (Project PI)
02/2023–01/2026 | \$300K | 3% Effort
- 9. Radiogenomics to Noninvasively Predict the Malignant Potential of Pancreatic IPMNs and Uncover Hidden Biology**
NIH/NCI R37 CA229810 (MERIT)

Rasool (Co-Investigator), Permuth, Jeong (MPIs)
09/2024–12/2026 | \$1.600M | 3% Effort

10. Patient-Reported Outcomes as Novel Biomarkers for Monitoring Cancer Progression and Treatment Response

Florida Department of Health, Florida Cancer Innovation Fund (FCIF)
Rasool (Co-Investigator), Jim, Brady, Schabath (MPIs)
04/2025–03/2026 | \$1.561M | 20% Effort

11. Harnessing Big Data to Understand the Confluence of HIV and Cancer Stigma in the US Southeast

NIH/NCI R01CA305812
Rasool (Co-Investigator), Islam (PI)
09/2025–08/2030 | \$4.072M | 5% Effort

12. Generating Accurate and Unbiased Radiogenomic Data Representation for Improved Clinical Care for Cancer Patients

University of Alabama at Birmingham (UAB)/Prime: PCORI
Rasool (Site PI)
08/2025–07/2028 | \$49,000 | 5% Effort

13. EDRN Collaborative Group Federated Learning Study for Lung Cancer Early Detection

NIH/NCI
Rasool (Co-Investigator, Schabath (PI))
04/2026–03/2027 | \$100,000 | 2.5% Effort

Pending Grant Applications (Total Count: 6)

1. Predicting Tertiary Lymphoid Structures with Artificial Intelligence to Inform Immunotherapy Response in NSCLC

Huntsman Cancer Institute (University of Utah) / Prime: NIH/NCI
Rasool (MPI), Stewart (Contact PI)
04/2026–03/2031 | \$1.861M | 25% Effort | Submitted: 06/2025

2. Federated Learning Enhancement Framework with Integrity, Provenance, and Authenticity of Data (FLIPAD)

Rochester Institute of Technology/Prime: NSF
Rasool (Site PI)
04/2026–03/2029 | \$100,000 | 5% Effort | Submitted: 01/2026

3. Quantitative Imaging Clinical Validation Center at Moffitt Cancer Center: Administrative Supplement-Privacy-Preserving AI for Early Lung Cancer Detection: A Federated Learning Demo

NIH/NCI
Rasool (Co-Investigator), Schabath and Heine (MPI)
04/2026–03/2027 | \$336,998 | 10% Effort | Submitted: 06/2025

4. Statewide Federated Learning Infrastructure for Early Lung Cancer Detection in Florida

FDOH Bankhead-Coley Cancer Research Program
Rasool (Contact PI)
07/2026–06/2029 | \$100,000 | 10% Effort | Submitted: 01/2026

5. **Collaborative Research: Trustable and Privacy-Preserving Federated Learning for Multi-Modal Medical Data Analysis**
NSF
Rasool (Site PI)
05/2026–04/2029 | \$100,000 | 10% Effort | Submitted: 08/2025
6. **A Multilevel Approach to Reduce Cancer in People at Risk for and with HIV**
American Cancer Society
Rasool (Co-Investigator), Vidrine (PI)
04/2026–03/2030 | \$4.069M | 5% Effort | Submitted: 12/2025

Completed Awards (Total Count: 21)

1. **Self-Assessment and Continual Learning on Edge Devices**
NSF OAC #2234836
Rasool (PI)
05/2020–08/2025 | \$500K | 15% Effort
2. **Multimodal AI Software for Cancer Research Data Commons**
NIH/NCI SBIR Phase I, Sub-award from IBIC Inc.
Rasool (Sub-award PI), IBIS Inc. (Prime)
08/2024–07/2025 | \$51,788 | 5% Effort
3. **Improving the Early Detection of Pancreatic Cancer by Developing a Multimodal Machine Learning Approach to Aid in Risk Stratification of Intraductal Papillary Mucinous Neoplasms (IPMNs)**
Cancer Epidemiology Team Science, Moffitt Cancer Center & Research Institute
Rasool (MPI), Permuth, Karolak (MPIs)
09/2024–08/2025 | \$34,000 | 0% Effort (in-kind)
4. **Enhancing Radiology Report Clarity, Conciseness, and Structure Using Local and Private Large Language Models (LLMs)**
Department of Diagnostic Imaging & Interventional Radiology, Moffitt Cancer Center & Research Institute
Rasool (Co-PI), Araujo (PI)
07/2024–06/2025 | \$18,500 | 0% Effort (in-kind)
5. **Enhancing Tissue Detection Accuracy in Digital Pathology through Uncertainty-Aware AI Methods**
Department of Pathology, Moffitt Cancer Center
Rasool (Co-PI), Bui (PI)
11/2024–06/2025 | \$10,000 | 0% Effort (in-kind)
6. **Leveraging Electronic Health Record Clinical Notes to Understand Cancer Care Equity Among People with HIV and Cancer at Moffitt Cancer Center**
Cancer Epidemiology Team Science, Moffitt Cancer Center
Rasool (MPI), Islam, Coghill (MPIs)
09/2024–08/2025 | \$18,000 | 0% Effort (in-kind)
7. **I-Corps: Detecting Performance Degradation and Failures of Deep Neural Networks in Cancer Imaging**

NSF Award #2304799

Rasool (PI)

03/2023–08/2024 | \$50,000 (Direct: \$45,000) | 6% Effort

8. **Rowan’s Prepare.AI — Rowan’s Graduate Fellowship Prepares for the Modern Age of AI**

U.S. Department of Education

Rasool (PI)

2018–2021 | \$1,250,000 | 10% Effort

9. **Trustworthy Artificial Intelligence for the Detection of COVID-19**

U.S. Department of Education

Rasool (PI)

2020–2021 | \$50,370 | 10% Effort

10. **Transforming Neuroimaging with Trustworthy and Reliable Artificial Intelligence**

Camden Health Initiative, Rowan University

Rasool (PI)

2020–2022 | \$100,000 | 5% Effort

11. **An Enhanced Bionic Limb**

New Jersey Health Foundation

Rasool (PI)

2018–2020 | \$35,000 | 5% Effort

12. **An Enhanced Bionic Limb — Next Steps**

New Jersey Health Foundation

Rasool (PI)

2019–2021 | \$35,000 | 5% Effort

13. **Uncertainty Estimation in Sequence Models with Bayesian Machine Learning**

Lockheed Martin

Rasool (PI)

2020–2021 | \$114,409 | 20% Effort

14. **Rotorcraft Landing Sites: Scaling and Generalizing the AI-Based Identification System**

U.S. Department of Transportation (via UTC/CAIT, Rutgers University)

Rasool (PI)

2021–2022 | \$60,000 | 10% Effort

15. **Rotorcraft Landing Sites: An AI-Based Identification System**

U.S. Department of Transportation (via UTC/CAIT, Rutgers University)

Rasool (PI)

2020–2021 | \$80,000 | 10% Effort

16. **Deep Phenotypic Characterization of Colorectal Cancer Patients to Identify Novel Predictors of Disease Recurrence**

NIH P30 CA076292-25 (Pilot Funding)

Rasool (Co-Investigator), Cleveland, Siegel, Kresovich, Stewart (PIs)

12/2023–11/2024 | \$114,976 | 3% Effort

17. **Moffitt Team Science Award (project title not specified)**

Moffitt Team Science Award, Moffitt Cancer Center

Rasool (Co-Investigator), Siegel (PI)
07/2023–06/2024 | \$150,000 | 3% Effort

18. **Developing Machine Learning Algorithms to Automatically Select Appropriate Images and Detect Body Composition Changes Using CT Scans**

HOB Innovation Funds, Moffitt Cancer Center

Rasool (Co-Investigator), Parker (PI)

09/2022–06/2023 | \$30,000 | 5% Effort

19. **Towards Safe and Reliable Autonomy in Sensor-Driven Systems (NSF #1903466)**

National Science Foundation

Rasool (Co-PI), Bouaynaya (PI)

2019–2022 | \$299,592 | 10% Effort

20. **Visualization of Rotorcraft Safety within a CAVE Virtual Reality Environment**

Federal Aviation Administration (FAA)

Rasool (Co-PI), Bouaynaya (PI)

2018–2022 | \$299,592 | 10% Effort

21. **AI-Based Virtual Reality Modeling for Identifying Threats to Water Resources in Landfills**

U.S. Department of Agriculture (USDA)

Rasool (Consultant; originally PI, awarded after move to Moffitt), Ahn (PI)

2022–2024 | \$299,592 | 10% Effort

Publications

Publications – Under Review (Total Count: 7)

1. Aakash Tripathi, Ian E Nielsen, Muhammad Umer, Ravi P Ramachandran, **Ghulam Rasool**, “Explainable AI in Genomics: Transcription Factor Binding Site Prediction with Mixture of Experts”, Under review. Preprint available: <https://arxiv.org/abs/2507.09754>.
2. Asim Waqas, Ehsan Ullah, Asma Khan, Farah Khalil, Zarifa Gahramanli Ozturk, Vaibhav Chumbalkar, Daryoush Saeed-Vafa, Zena Jameel, Wei-Shen Chen, Humberto Trejo Bittar, Jasreman Dhillon, Rajendra S. Singh, Andrey Bychkov, Anil V. Parwani, Marilyn M. Bui, Matthew B. Schabath, **Ghulam Rasool**, “Reasoning Beyond Accuracy: Expert Evaluation of Large Language Models in Diagnostic Pathology,” under review.
Preprint: <https://www.medrxiv.org/content/10.1101/2025.04.11.25325686v2>.
3. Sabeen Ahmed, Nathan Parker, Margaret Park, Daniel Jeong, Lauren Peres, Evan W. Davis, Jennifer B. Permuth, Erin Siegel, Matthew B. Schabath, Yasin Yilmaz, and **Ghulam Rasool**, “Multimodal AI-driven Biomarker for Early Detection of Cancer Cachexia,” PMID: [PMC12458964](https://pubmed.ncbi.nlm.nih.gov/PMC12458964/), Preprint available: <https://doi.org/10.1101/2025.09.17.25336028>.
4. Sabeen Ahmed, Nathan Parker, Margaret Park, Daniel Jeong, Lauren Peres, Evan W. Davis, Jennifer B. Permuth, Erin Siegel, Matthew B. Schabath, Yasin Yilmaz, and **Ghulam Rasool**, “Reliable Radiologic Skeletal Muscle Area Assessment – A Biomarker for Cancer Cachexia Diagnosis,” PMID: [PMC12045449](https://pubmed.ncbi.nlm.nih.gov/PMC12045449/), Preprint available: <https://www.medrxiv.org/content/10.1101/2025.04.21.25326162v1>.
5. Sofia A. Haddan, Asim Waqas, **Ghulam Rasool**, Matthew B. Schabath, “Personalized risk models based on lung cancer screening history: Post hoc analysis of the National Lung Cancer Screening Trial (NLST),” under review.

6. Iryna Hartsock, Nikolas Koutsoubis, Sabeen Ahmed, Nathan Parker, Matthew Schabath, Cyrillo Araujo, Aliya Qayyum, Cesar Lam, Robert Gatenby, and **Ghulam Rasool**, “Clinical AI in Radiology: Foundations, Trends, and Emerging Directions with Use Cases from Moffitt Cancer Center”, under review in MDPI Cancers. Preprint available: <https://www.preprints.org/manuscript/202602.0555>
7. Evan Davis, Margaret Park, Toni Basinski, Solomon Alhassan, Maria Gomez, Maria Genilo-Delgado, Andrew Sinnamon, Pamela Hodul, Aleksandra Karolak, Zena Sayegh, Jonathan Nguyen, Brittany Rumens, Jiannong Li, Aakash Tripathi, Nathan Parker, Jose Pimiento, **Ghulam Rasool**, Alexandra Tassielli, Dung-Tsa Chen, Barbara Centeno, Kun Jiang, Daniel Jeong, and Jennifer Permeth, “Radiomic Signature Improves Preoperative Risk Stratification Beyond International Consensus Guidelines in Intraductal Papillary Mucinous Neoplasms”, under review in Cancer Research Communications.

Peer-Reviewed Journal and Conference Publications (Total Count: 79)

2026

1. Jacob R Epifano, Stephen Glass, Ravi P Ramachandran, Sharad Patel, Aaron J Masino, and **Ghulam Rasool**, “Deployment of a Robust and Explainable Mortality Prediction Model: The COVID-19 Pandemic and Beyond”, accepted for publication in SmartHealth.
2. Giuseppina Carannante, Nidhal Bouaynaya, Dimah Dera, Hassan Fathallah-Shaykh, **Ghulam Rasool**, “Testing the Trust: Verification and Validation of Bayesian Segmentation under Uncertainty. Accepted for publication as a validation paper in Medical Imaging with Deep Learning (MIDL), Taipei, 8-10 July 2026. Accepted version available: <https://openreview.net/forum?id=Dl5KEuccye>.
3. Aakash Tripathi, Asim Waqas, Kavya Venkatesan, Ehsan Ullah, Asma Khan, Farah Khalil, Zarifa Gahramanli Ozturk, Daryoush Saeed-Vafa, Wei-Shen Chen, Marilyn M. Bui, Matthew B. Schabath, **Ghulam Rasool**, “Using Consensus-based Reasoning and Large Language Models to Extract Structured Data from Surgical Pathology Reports,” Laboratory Investigation, Article 104272, Volume 106, Issue 2, Preprint available: <https://doi.org/10.1016/j.labinv.2025.104272>.
4. Asim Waqas, **Ghulam Rasool**, and Matthew B. Schabath, “Trans-omics Gene-Smoking Interactions: A New Framework for Stratifying Smoking-Related Lung Cancer Risk” in Press, American Journal of Respiratory and Critical Care Medicine (AJRCCM).

2025

5. Oya Altinok, **Ghulam Rasool**, Asim Waqas, Matthew B. Schabath, Albert Guvenis, “Habitat Radiomics Predict HPV Status in Oropharyngeal Cancer”, Cancer Medicine, Volume 14, Issue 24, <https://doi.org/10.1002/cam4.71481>
6. Bradley Gathers, Ian E. Nielsen, Keith W. Soules, Ozan Tekben, Ravi P. Ramachandran, Nidhal C. Bouaynaya, Hassan M. Fathallah-Shaykh, and **Ghulam Rasool**, “IntActEval: A Continuous Framework for Quantitatively Evaluating Faithfulness of Feature Visualization Methods”, IEEE Access, Vol. 13, pp. 196766–196782. <https://doi.org/10.1109/ACCESS.2025.3631378>
7. Jennifer Permeth, Margaret Park, Evan Davis, Solomon Alhassan, J. Arnoletti, Toni Basinski, Ashley McKee, Mark Bloomston, Tiffany Carson, Tiago Biachi de Castria, Dung-Tsa Chen, Elena Cortizas, Sylvia Crowder, Maria Genilo Delgado, Wade Douglas, Jason Fleming, Pamela Hodul, Kevin Huguet, Kun Jiang, Dae Won Kim, John Koomen, Anjuli Luthra, Mokenge Malafa, Anjana Menon, Raiza Morales, Nipun Merchant, Kenneth Meredith, Qianxing Mo, Manual Molina-Vega, Lina Moreno-Urazan, Kayode Olumoyin, Nathan Parker, Jose Pimiento, **Ghulam Rasool**, Katarzyna Rejniak, Samer Sansil,

Lauren Sparks, Paul Stewart, Alexandra Tassielli, Jamie Teer, Dan Viet Tran, Jose Trevino, Vic Velanovich, Christopher Whelan, Daniel Jeong, Sarah Judge, and Andrew Judge, “Race-based Differences in Serum Biomarkers for Cancer-Associated Cachexia in a Diverse Cohort of Patients with Pancreatic Ductal Adenocarcinoma,” *Commun Med* 6, 19 (2026).
Available: <https://doi.org/10.1038/s43856-025-01277-9>.

8. Giuseppina Carannante, Nidhal Bouaynaya, Dimah Dera, Hassan M. Fathallah-Shaykh, and **Ghulam Rasool**, “SUPER-Net: Trustworthy Image Segmentation via Uncertainty Propagation in Encoder-Decoder Networks,” *Pattern Recognition*, 172, 112503.
<https://doi.org/10.1016/j.patcog.2025.112503>.
9. Dominic Flack, Aakash Tripathi, Asim Waqas, **Ghulam Rasool**, and Dimah Dera, “Robust Multimodal Fusion for Oncology,” *Cancer Informatics*. <https://doi.org/10.1177/11769351251376192>
10. Aakash Tripathi, Asim Waqas, Yasin Yilmaz, and **Ghulam Rasool**, “HONeYBEE: Enabling Scalable Multimodal AI in Oncology Through Foundation Model–Driven Embeddings,” *npj Digital Medicine*, 8, 622 (2025). <https://doi.org/10.1038/s41746-025-02003-4>.
11. Asim Waqas, Aakash Tripathi, Sabeen Ahmed, Ashwin Mukund, Hamza Farooq, Matthew B. Schabath, Paul Stewart, Mia Naeini, and **Ghulam Rasool**, “Self-Normalizing Multi-Omics Neural Network for Pan-Cancer Prognostication,” *International Journal of Molecular Sciences*, 2025, 26(15), 7358.
<https://doi.org/10.3390/ijms26157358>.
12. Evan W Davis, Margaret A Park, Toni L Basinski, J Pablo Arnoletti, Mark Bloomston, Tiffany L Carson, Tiago Biachi De Castria, Dung-Tsa Chen, Elena M Cortizas, Sylvia L Crowder, Maria Genilo-Delgado, Wade G Douglas, Kevin L Huguet, Kun Jiang, Pamela J Hodul, Aleksandra Karolak, Dae Won Kim, John M Koomen, Anjana A Menon, Qianxing Mo, Shaffer R Mok, Manuel A Molina-Vega, Lina Moreno-Urazan, Sabeen Ahmed, Nathan H Parker, Jose M Pimiento, **Ghulam Rasool**, Lauren M Sparks, Paul A Stewart, Alexandra F Tassielli, Jamie K Teer, Jose G Trevino, Vic Velanovich, Xuefeng Wang, Christopher J Whelan, Sarah M Judge, Andrew R Judge, Jason B Fleming, Mokenge P Malafa, Daniel Jeong, Jennifer B Permuth, “Impact of Edema on Measured Muscle Change During Pancreatic Ductal Adenocarcinoma (PDAC) Treatment,” *AACR Journal of Cancer Epidemiology, Biomarkers, and Prevention*, <https://doi.org/10.1158/1055-9965.epi-25-0237>.
13. Mana Moassefi, Sina Houshmand, Shahriar Faghani, Peter Y. Chang, Shawn Sun, Katherine P. Andriole, Judy W. Gichoya, Neil Bhatia, Aakash Tripathi, **Ghulam Rasool**, Les Folio, and Bradley J. Erickson, “Cross-Institutional Evaluation of Large Language Models for Radiology Diagnosis Extrac-tion: A Prompt-Engineering Perspective,” *Journal of Imaging Informatics in Medicine*.
<https://doi.org/10.1007/s10278-025-01523-5>.
14. Iryna Hartsock, Cyrillo Araujo, Les Folio, and **Ghulam Rasool**, “Improving Radiology Report Con-ciseness and Structure via Local Large Language Model,” *Journal of Imaging Informatics in Medicine* (2025). <https://doi.org/10.1007/s10278-025-01510-w>.
15. Nikolas Koutsoubis, Yasin Yilmaz, Matthew Schabath, **Ghulam Rasool**, “Privacy-Preserving Feder-ated Learning and Uncertainty Quantification in Medical Imaging,” *RSNA Radiology AI*.
<https://pubs.rsna.org/doi/10.1148/ryai.240637>.
16. Ehsan Ullah, Mirza Mansoor Baig, Asim Waqas, **Ghulam Rasool**, Rajendra Singh, Hamid Gho-lam Hossieni, and Anil V. Parwani, “Multimodal Generative AI for Anatomic Pathology – A Review of Current Applications to Envisage the Future Direction,” *Advances in Anatomic Pathology* (2025).
<https://pubmed.ncbi.nlm.nih.gov/40298358/>.
17. Victor Brodsky, Ehsan Ullah, Andrey Bychkov, Andrew H. Song, Eric E. Walk, Peter Louis, **Ghu-lam Rasool**, Rajendra S. Singh, Faisal Mahmood, Marilyn M. Bui, Anil V. Parwani, “Generative

Artificial Intelligence in Anatomic Pathology,” *Archives of Pathology & Laboratory Medicine* (2025). <https://doi.org/10.5858/arpa.2024-0215-RA>.

2024

18. Jacob Epifano, T. Duong, Ravi Ramachandran, and **Ghulam Rasool**, “Efficient Scaling of Bayesian Neural Networks,” *IEEE Access*, Volume 12, pp. 150953–150961. <https://doi.org/10.1109/ACCESS.2024.3479913>.
19. Aakash Tripathi, Asim Waqas, Kavya Venkatesan, Yasin Yilmaz, **Ghulam Rasool**, “Building Flexible, Scalable, and Machine Learning-ready Multimodal Oncology Datasets,” *Sensors*, 2024, 24(5), 1634. <https://doi.org/10.3390/s24051634>.
20. Hikmat Khan, Nidhal Bouaynaya, **Ghulam Rasool**, “Brain-Inspired Continual Learning: Robust Feature Distillation and Re-consolidation for Class Incremental Learning,” *IEEE Access*. <https://doi.org/10.1109/ACCESS.2024.3369488>.
21. Hikmat Khan, Nidhal Bouaynaya, and **Ghulam Rasool**, “Adversarially Diversified Rehearsal Memory (ADRM): Mitigating Memory Overfitting Challenge in Continual Learning,” accepted for publication at *IEEE IJCNN/WCCI 2024*, Yokohama, Japan, 30 June – 5 July 2024.
22. Margaret A. Park, Christopher J. Whelan, Sabeen Ahmed, Tabitha Boeringer, Joel Brown, Sylvia L. Crowder, Kenneth Gage, Christopher Gregg, Daniel K. Jeong, Heather S.L. Jim, Andrew R. Judge, Tina M. Mason, Nathan Parker, Smitha Pillai, Aliya Qayyum, Sahana Rajasekhara, **Ghulam Rasool**, Sara M. Tinsley, Matthew B. Schabath, Paul Stewart, Jeffrey West, Patricia McDonald, and Jennifer B. Permuth, “Defining and Addressing Research Priorities in Cancer Cachexia through Transdisciplinary Collaboration,” *Cancers*, 2024, 16(13), 2364. <https://doi.org/10.3390/cancers16132364>.
23. Spyridon Bakas, Philipp Vollmuth, Norbert Galldiks, Thomas C. Booth, Hugo J. W. L. Aerts, Wenya Linda Bi, Benedikt Wiestler, Pallavi Tiwari, Sarthak Pati, Ujjwal Baid, Evan Calabrese, Philipp Lohmann, Martha Nowosielski, Rajan Jain, Rivka Colen, Marwa Ismail, **Ghulam Rasool**, Janine M. Lupo, Hamed Akbari, Joerg C. Tonn, David Macdonald, Michael Vogelbaum, Susan M. Chang, Christos Davatzikos, Javier E. Villanueva-Meyer, and Raymond Y. Huang, “Artificial Intelligence for Response Assessment in Neuro Oncology (AI-RANO), Part 2: Recommendations for Standardization, Validation, and Good Clinical Practice,” *The Lancet Oncology*, Volume 25, Issue 11, e589–e601. [https://doi.org/10.1016/S1470-2045\(24\)00315-2](https://doi.org/10.1016/S1470-2045(24)00315-2).
24. Giuseppina Carannante, Nidhal Bouaynaya, **Ghulam Rasool**, and Lyudmila Mihaylova, “BaSIS-Net: From Point Estimate to Predictive Distribution in Neural Networks – A Bayesian Sequential Importance Sampling Framework,” *Transactions on Machine Learning Research (TMLR)*. <https://openreview.net/forum?id=V92PnXQ7UW>.
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Peer-Reviewed Scientific Abstracts with Oral or Poster Presentations (Total Count: 80)

1. Hanieh Ajami, Asim Waqas, John Templeton, Matthew B. Schabath, and **Ghulam Rasool**, “Enhancing lung cancer risk prediction with longitudinal fusion of LDCT data”, accepted for oral presentation at the 2026 Annual American Association for Cancer Research (AACR) Meeting, San Diego, California, April 17-22, 2026.
2. Oya Altinok, **Ghulam Rasool**, Asim Waqas, Matthew B. Schabath, Albert Guvenis, “AI Algorithms to Predict HPV Status from Computed Tomography in Oropharyngeal Cancer”, accepted for poster presentation at the 2026 Annual American Association for Cancer Research (AACR) Meeting, San Diego, California, April 17-22, 2026.
3. Aakash Tripathi, Asim Waqas, Evan Davis, Jennifer Pernuth, Joaquim Farinhas, Yasin Yilmaz, Matthew B. Schabath, and **Ghulam Rasool**, “Real-world evaluation of multimodal AI: Foundation model-driven multimodal AI for GBM, NSCLC, and PDAC”, accepted for poster presentation at the 2026 Annual American Association for Cancer Research (AACR) Meeting, San Diego, California, April 17-22, 2026.
4. Asim Waqas, Aakash Tripathi, Kris Bowles, Brianna Miner, Jessica Y. Islam, Anna E. Coghill1, Anastasia Jones, Matthew B. Schabath, and **Ghulam Rasool**, “Multi-agent AI orchestration for temporal-aware extraction of social determinants of health from unstructured clinical records in cancer populations”, accepted for poster presentation at the 2026 Annual American Association for Cancer Research (AACR) Meeting, San Diego, California, April 17-22, 2026.

5. Aakash Tripathi, Osama Elzaafarany, Sepideh Mokhtari, Michael Vogelbaum, Marilyn Bui, and **Ghulam Rasool**, “Multi-Agent System for Automated Extraction of Neuro-Oncology Biomarkers from Pathology Reports”, accepted for presentation, the United States and Canadian Academy of Pathology’s (USCAP) 115th Annual Meeting, March 21–26, 2026, San Antonio, TX.
6. Hanieh Ajami, Asim Waqas, John Michael Templeton, Matthew B. Schabath, Ghulam Rasool, and **Ghulam Rasool**, “Enhancing Lung Cancer Risk Prediction with Longitudinal Fusion of LDCT Data”, Poster Presentation by Hanieh Ajami, 2025 Robert Gillies Machine Learning Workshop, October 30–31, 2025, Tampa, FL.
7. Aakash Tripathi, Asim Waqas, Jessica Islam, Kristina Bowles, Brianna Miner, Anna Coghill, Anastasia Jones, Matthew Schabath, and **Ghulam Rasool**, “CLEVER Multi-Agent AI Orchestration for Temporal-Aware Clinical Variable Extraction from Unstructured Medical Records”, Oral Presentation by Aakash Tripathi and Asim Waqas, 2025 Robert Gillies Machine Learning Workshop, October 30–31, 2025, Tampa, FL.
8. Sabeen Ahmed, Nathan Parker, Margaret Park, Evan W. Davis, Jennifer B. Permuth, Matthew B. Schabath, Yasin Yilmaz, and **Ghulam Rasool**, “A Multimodal AI Biomarker for Early Detection of Cancer Cachexia Integrating Clinical, Radiologic, and Laboratory Indicators”, Oral Presentation by Sabeen Ahmed, 2025 Robert Gillies Machine Learning Workshop, October 30–31, 2025, Tampa, FL.
9. Iryna Hartsock, Cyrillo Araujo, Cesar Lam, Les Folio, and **Ghulam Rasool**, “Improving Radiology Reporting with Local Reasoning Large Language Models”, Oral Presentation by Iryna Hartsock, 2025 Robert Gillies Machine Learning Workshop, October 30–31, 2025, Tampa, FL.
10. Asim Waqas, Aakash Tripathi, Matthew B. Schabath, Yasin Yilmaz **Ghulam Rasool**, “An End-to-End Multimodal AI Framework for Precision Oncology - From Data Integration to Clinical Decision Support”, Poster Presentation Asim Waqas, 2025 Robert Gillies Machine Learning Workshop, October 30–31, 2025, Tampa, FL.
11. Sabeen Ahmed, Nathan Parker, Margaret Park, Evan W. Davis, Jennifer B. Permuth, Matthew B. Schabath, Yasin Yilmaz, and **Ghulam Rasool**, “A Multimodal AI Biomarker for Early Detection of Cancer Cachexia Integrating Clinical, Radiologic, and Laboratory Indicators”, Early Detection of Cancer Conference, October 21–23, 2025, Portland, OR.
12. Hanieh Ajami, John Templeton, Matthew Schabath, and **Ghulam Rasool**, “HybridSybil: Integrating 3D-CNN and Vision Transformer for Longitudinal Lung Cancer Risk Prediction from LDCT”, Oral and poster presentations at the Early Detection of Cancer Conference, October 21–23, 2025, Portland, OR.
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15. Aleksandra Karolak, Evan W. Davis, Mouktik Isukapalli, Rohit Veligeti, Margaret A. Park, Jamie K. Teer, Daniel K. Jeong, Kun Jiang, Dung-Tsa Chen, Jennifer B. Permuth, and **Ghulam Rasool**, “Abstract A043: Mutational Profiling and Machine Learning for Risk Stratification and Biomarker Identification in Intraductal Papillary Mucinous Neoplasms Progressing to Pancreatic Cancer,” AACR Special

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17. Monique Shotande, Jude Delikat, K. Ruwani Fernando, **Ghulam Rasool**, Taylor Welniak, Jacque-line Andreozzi, Peter Johnstone, Evangelia Katsoulakis, Adam Dicker, Heather Jim, and Issam Naqa, “Enhancing Adverse Event Monitoring in Prostate Cancer Care with Large Language Models,” poster presentation at the 2025 American Society for Radiation Oncology (ASTRO) Annual Meeting.
18. Evan W. Davis, Aakash Tripathi, Toni L. Basinski, Kun Jiang, Dung-Tsa Chen, Daniel Jeong, Jennifer B. Permuth, and **Ghulam Rasool**, “Applying Large Language Models for Clinical Data Abstraction in Patients with Pancreatic Tumors,” podium talk at the 2025 Moffitt Scientific Symposium, May 13, 2025, Tampa, Florida.
19. Sabeen Ahmed, Nathan Parker, Margaret Park, Daniel Jeong, Lauren Peres, Evan W. Davis, Jennifer B. Permuth, Erin Siegel, Matthew B. Schabath, Yasin Yilmaz, and **Ghulam Rasool**, “Multimodal AI for Accurate Prediction of Cancer Cachexia,” podium talk at the 2025 Moffitt Scientific Symposium, May 13, 2025, Tampa, Florida.
20. Kavya Venkatesan, Asim Waqas, Natasha Avrilien, Taylor Welniak, Yvelise Rodriguez, Hirut G. Kas-saye, Daniel Parras, Joycelyn Lawrence, Saint Anthony Amofah, Katherine Chung-Bridges, Brian Gon-zalez, Jennifer Vidrine, and **Ghulam Rasool**, “Evaluating the Safety and Reliability of AI Chatbots for Cancer Information – A Partnership Between Health Choice Network (HCN) and Moffitt Cancer Center,” poster presentation at the 2025 Moffitt Scientific Symposium, May 13, 2025, Tampa, Florida.
21. Aakash Tripathi, Asim Waqas, Matthew B. Schabath, Yasin Yilmaz, and **Ghulam Rasool**, “HONeY-BEE: Enabling Scalable Multimodal AI in Oncology Through Foundation Model–Driven Embeddings,” poster presentation at the 2025 Moffitt Scientific Symposium, May 13, 2025, Tampa, Florida.
22. Asim Waqas, Ehsan Ullah, Asma Khan, Farah Khalil, Zarifa Gahramanli Ozturk, Vaibhav Chum-balkar, Daryoush Saeed-Vafa, Zena Jameel, Wei-Shen Chen, Humberto Trejo Bittar, Jasreman Dhillon, Rajendra S. Singh, Andrey Bychkov, Anil V. Parwani, Marilyn M. Bui, Matthew B. Schabath, and **Ghulam Rasool**, “Exploring Reasoning Capabilities of Large Language Models (LLMs) in Pathol-ogy,” poster presentation at the 2025 Moffitt Scientific Symposium, May 13, 2025, Tampa, Florida.
23. Nikolas Koutsoubis, Asim Waqas, John Heine, Matthew B. Schabath, and **Ghulam Rasool**, “Harness-ing Distributed Intelligence with Privacy Preservation: Federated Learning for the Early Detection of Lung Cancer,” poster presentation at the 2025 Moffitt Scientific Symposium, May 13, 2025, Tampa, Florida.
24. Aakash Tripathi and **Ghulam Rasool**, “HoneyBee: A Scalable, Trustworthy, and Modular Framework for Creating Multimodal Oncology Datasets and Models,” oral presentation at the inaugural National AI Research Resource (NAIRR) meeting, Feb 19–21, 2025, Arlington, VA, USA.
25. **Ghulam Rasool** and Aakash Tripathi, “The Blue Scrubs: An Open-Source Platform for Evaluating Generative AI in Healthcare Applications,” oral presentation at the inaugural NAIRR meeting, Feb 19–21, 2025, Arlington, VA, USA.

26. Asim Waqas, Aakash Tripathi, Mia Naeini, Paul A. Stewart, Matthew B. Schabath, and **Ghulam Rasool**, “Using Patient Embeddings from Foundation Models for Enhanced Survival Analysis in Lung Squamous Cell Carcinoma,” oral presentation at the American Thoracic Society (ATS) Annual Meeting, May 16–21, 2025, San Francisco, CA.
<https://doi.org/10.1164/ajrccm.2025.211.Abstracts.A3108>
27. Asim Waqas, Aakash Tripathi, Mia Naeini, Paul A. Stewart, Matthew B. Schabath, and **Ghulam Rasool**, “Abstract 991: PARADIGM: an embeddings-based multimodal learning framework with foundation models and graph neural networks,” Poster presentation at the 2025 AACR Annual Meeting, April 25–30, 2025, Chicago, IL. <https://doi.org/10.1158/1538-7445.AM2025-991>
28. Evan W. Davis, Margaret A. Park, Toni L. Basinski, J. Pablo Arnoletti, Francisca Beato, Mark Bloomston, Tiffany L. Carson, Tiago Biachi de Castria, Dung-Tsa Chen, Elena M. Cortizas, Sylvia L. Crowder, Maria Genilo-Delgado, Wade G. Douglas, Kevin L. Huguet, Kun Jiang, Pamela J. Hodul, Aleksandra Karolak, Dae Won Kim, John Koomen, Anjana A. Menon, Kenneth Meredith, Qianxing Mo, Shaffer R. Mok, Lina Moreno-Urazan, Sabeen Ahmed, Nathan Parker, Jose M. Pimiento, **Ghulam Rasool**, Lauren M. Sparks, Paul Stewart, Alexandra F. Tassielli, Jamie K. Teer, Dan Viet Tran, Jose G. Trevino, Vic Velanovich, Xuefeng Wang, Christopher J. Whelan, Sarah M. Judge, Andrew R. Judge, Jason B. Fleming, Daniel Jeong, and Jennifer B. Permuth, “Abstract 4904: Say it ain’t psoas: Addressing edema in pancreatic cancer body composition analyses,” poster presentation at the 2025 AACR Annual Meeting, April 25–30, 2025, Chicago, IL. <https://doi.org/10.1158/1538-7445.AM2025-4904>.
29. Sofia A. Haddan, Matthew B. Schabath, Asim Waqas, and **Ghulam Rasool**, “Abstract 6211: Personalized risk models based on lung cancer screening history: post hoc analysis of the National Lung Screening Trial (NLST),” poster presentation at the 2025 AACR Annual Meeting, April 25–30, 2025, Chicago, IL. <https://doi.org/10.1158/1538-7445.AM2025-6211>
30. **Ghulam Rasool**, Iryna Hartsock, Cyrillo Araujo, and Les Folio, “Improving Radiology Report Conciseness and Structure via Locally Run Large Language Models,” oral presentation at the 2025 SIIM Annual Meeting, May 21–23, 2025, Portland, OR.
<https://link.springer.com/article/10.1007/s10278-025-01501-x>
31. Aakash Tripathi, Asim Waqas, Yasin Yilmaz, Matthew B. Schabath, and **Ghulam Rasool**, “Abstract 3641: Predicting treatment outcomes using cross-modality correlations in multimodal oncology data,” poster presentation at the 2025 AACR Annual Meeting, April 25–30, 2025, Chicago, IL. <https://doi.org/10.1158/1538-7445.AM2025-3641>
32. Sabeen Ahmed, Nathan Parker, Margaret Park, Daniel Jeong, Lauren Peres, Evan W. Davis, Jennifer B. Permuth, Erin Siegel, Matthew B. Schabath, Yasin Yilmaz, and **Ghulam Rasool**, “Abstract 1143: Reliable skeletal muscle area quantification and clinical data integration for predicting cancer cachexia,” oral presentation at the 2025 AACR Annual Meeting, April 25–30, 2025, Chicago, IL. <https://doi.org/10.1158/1538-7445.AM2025-1143>
33. **Ghulam Rasool**, Manijeh Berenji, and Les Folio, “Automated Occupational Lung Disease Identification, Characterization, Quantification and Localization on Chest X-Rays,” poster presentation at the AMIA 2025 Informatics Summit, March 10–13, Pittsburgh, PA.
34. Nikolas Koutsoubis, Asim Waqas, Matthew B. Schabath, and **Ghulam Rasool**, “Harnessing Distributed Intelligence with Privacy Preservation: Federated Learning for the Early Detection of Lung Cancer,” poster presentation at NVIDIA GTC 2025, March 17–21, San Jose, USA.
35. Sabeen Ahmed, Nathan Parker, and **Ghulam Rasool**, “Reliable Radiographic Skeletal Muscle Area Assessment: A Biomarker for Cancer Cachexia Diagnosis,” poster presentation at NVIDIA GTC 2025, March 17–21, San Jose, USA.

36. Aakash Tripathi and **Ghulam Rasool**, “Accelerate Cancer Research with AI-Driven Multimodal Data Integration,” poster presentation at NVIDIA GTC 2025, March 17–21, San Jose, USA.
37. Aakash Tripathi, Asim Waqas, Kavya Venkatesan, Ehsan Ullah, Marilyn Bui, and **Ghulam Rasool**, “AI-Driven Extraction of Key Clinical Data from Pathology Reports to Enhance Cancer Registries,” accepted for publication in the United States and Canadian Academy of Pathology (USCAP) 114th Annual Meeting, March 22–27, 2025, Boston, MA.
<https://doi.org/10.1016/j.labinv.2024.103629>.
38. Hongzhi Xu, Hy Nguyen, and **Ghulam Rasool**, “A Machine Learning-Assisted Automated Digital Urine Cytology Screening System,” accepted for publication in the USCAP 114th Annual Meeting, March 22–27, 2025, Boston, MA.
<https://www.sciencedirect.com/science/article/pii/S0023683724023134>.

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39. Mahsa Chitsaz, **Ghulam Rasool**, Hy Nguyen, Jasreman Dhillon, and Hongzhi Xu, “Developing an Automated Urine Cytology Diagnostic Workflow with The Paris System (TPS)–Based Machine Learning (ML) Model,” oral presentation at the American Society of Cytopathology 2024.
<https://www.sciencedirect.com/science/article/pii/S2213294524000796>.
40. Iryna Hartsock and **Ghulam Rasool**, “Improving Radiology Report Conciseness and Structure via Open-Source Local Large Language Models,” Second USF Artificial Intelligence + X Symposium, October 25, 2024, Tampa, FL.
41. Sabeen Ahmed, Nathan Parker, Daniel Jeong, Margaret Park, Lauren Peres, Jennifer Permeth, Matthew Schabath, Yasin Yilmaz, and **Ghulam Rasool**, “Reliable Radiographic Skeletal Muscle Area Assessment: A Biomarker for Cachexia Diagnosis,” Second USF Artificial Intelligence + X Symposium, October 25, 2024, Tampa, FL.
42. Aakash Tripathi, Asim Waqas, Yasin Yilmaz, and **Ghulam Rasool**, “Advancing Cancer Research Through Integrated Multimodal Data Analysis: A Comprehensive Framework for Precision Oncology,” Second USF Artificial Intelligence + X Symposium, October 25, 2024, Tampa, FL.
43. Asim Waqas, Aakash Tripathi, and **Ghulam Rasool**, “Local and Private Large Language Models (LLMs) for Cancer Registry – An Automated Tool for Cancer Researchers, Clinicians, and Registrars,” Second USF Artificial Intelligence + X Symposium, October 25, 2024, Tampa, FL.
44. Nikolas Koutsoubis, Yasin Yilmaz, and **Ghulam Rasool**, “Harnessing Distributed Intelligence with Privacy Preservation: Federated Learning in Oncology – A Preliminary Simulation Study,” Second USF Artificial Intelligence + X Symposium, October 25, 2024, Tampa, FL.
45. Cyrillo Araujo, Iryna Hartsock, Les Folio, and **Ghulam Rasool**, “Elevating Report Quality and Clarity Using Large Language Models,” oral presentation at the 49th Annual Meeting of Society for Advanced Body Imaging (SABI), Washington, DC, October 5–9, 2024.
46. Les Folio and **Ghulam Rasool**, “Improving Radiologist Report Conciseness and Structure Leveraging Large Language Models (LLMs),” workshop (co-presenter with Les Folio) at the 49th Annual Meeting of SABI, Washington, DC, October 5–9, 2024.
47. **Ghulam Rasool**, Asim Waqas, Aakash Tripathi, Ehsan Ullah, and Marilyn Bui, “Extraction of Discrete Information from Pathology Reports Using Local and Private LLMs,” oral presentation at Pathology Visions 2024, Orlando, Nov 3–5, 2024.
<https://www.sciencedirect.com/science/article/pii/S215335392500001X>.

48. Victor Brodsky, Ehsan Ullah, Andrey Bychkov, Andrew H. Song, Eric Walk, Peter Louis, **Ghulam Rasool**, Raj Singh, Faisal Mahmood, Marilyn M. Bui, and Anil V. Parwani, “Generative AI in Anatomical Pathology: Today’s Innovations and Tomorrow’s Possibilities,” oral presentation at Pathology Visions 2024 (presented by Dr. Ehsan Ullah), Orlando, Nov 3–5, 2024.
<https://www.sciencedirect.com/science/article/pii/S215335392500001X>.
49. **Ghulam Rasool**, “Developing Representative Multimodal Embeddings for Foundation Models in Oncology,” oral presentation at Mayo AI Summit 2024: Generative and Multimodal AI – Potentials and Challenges, July 8–9, 2024, Rochester, Minnesota, USA.
50. Nikolas Koutsoubis, Asim Waqas, Matthew Schabath, and **Ghulam Rasool**, “Harnessing Distributed Intelligence with Privacy Preservation: Federated Learning in Oncology – A Preliminary Simulation Study,” oral presentation at the Cancer Biomarkers AI and Bioinformatics Workshop, August 13–15, 2024, California Institute of Technology, Pasadena, CA.
51. Aakash Tripathi, Asim Waqas, Yasin Yilmaz, and **Ghulam Rasool**, “Unifying Multimodal Data, Time Series Analytics, and Contextual Medical Memory: Introducing MINDS as an Oncology-Centric Cloud-Based Platform,” poster presentation at the 2024 NCCN Annual Conference, April 5–6, 2024.
52. Sabeen Ahmed, Nathan Parker, and **Ghulam Rasool**, “Early Diagnosis of Cancer Cachexia Using Body Composition Index as the Radiographic Biomarker,” poster presentation at the 2024 NCCN Annual Conference, April 5–6, 2024.
53. Asim Waqas, Aakash Tripathi, Sabeen Ahmed, Paul Stewart, Hamza Farooq, and **Ghulam Rasool**, “SeNMo: A Self-Normalizing Deep Learning Model for Enhanced Multi-Omics Data Analysis in Oncology,” poster presentation at the AACR Annual Meeting 2024. [Abstract](#).
54. Aakash Tripathi, Asim Waqas, Yasin Yilmaz, and **Ghulam Rasool**, “Multimodal Transformer Model Improves Survival Prediction in Lung Cancer Compared to Unimodal Approaches,” poster presentation at the AACR Annual Meeting 2024. [Abstract](#).
55. Sabeen Ahmed, Nathan Parker, and **Ghulam Rasool**, “Early Diagnosis of Cancer Cachexia Using Body Composition Index as the Radiographic Biomarker,” poster presentation at NVIDIA GTC 2024, San Jose, March 18–22, 2024.
56. Asim Waqas, Aakash Tripathi, Sabeen Ahmed, Paul Stewart, and **Ghulam Rasool**, “Hierarchical Multimodal Learning on Pan-Squamous Cell Carcinomas for Improved Survival Outcomes,” poster presentation at the 2024 NCCN Annual Conference, April 5–6, 2024.

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57. Asim Waqas, Aakash Tripathi, Ashwin Mukund, Paul Stewart, Mia Naeini, and **Ghulam Rasool**, “Hierarchical Multimodal Learning on Pan-Squamous Cell Carcinomas for Improved Survival Outcomes,” poster presentation at the AI+X Symposium, University of South Florida, September 29, 2023.
58. **Ghulam Rasool** and Les Folio, “Towards Patient Consumable Radiology Reports: Improving Content Signal-to-Noise Ratio (SNR) While Converting Medical Jargon to Plain English via GPT-4,” presentation at the Radiological Society of North America (RSNA) Annual Meeting, Chicago, November 2023. Highlighted in RSNA Daily Bulletin:
<https://dailybulletin.rsna.org/db23/index.cfm?pg=23thu08>.
59. Joanna J. Song, Harshna V. Vadvala, Sabeen Ahmed, Sabrina Khan, Lucas S. Folio, **Ghulam Rasool**, João Santinha, and Les R. Folio, “Innovative Approaches to Tumor Volume Quan-

tification: An Imaging Informatics Hackathon Challenge,” poster presented at the 2023 Annual Meeting of the Florida Radiological Society (FRS).

60. Asim Waqas, Paul Stewart, Hamza Farooq, and **Ghulam Rasool**, “Integrative Relational Learning on Multimodal Cancer Data for Improved Clinical Predictions,” poster presentation at MCBIOS 2023 Conference, University of Dallas, Dallas, TX, March 15–17, 2023.

2022 and Earlier

61. Giuseppina Carannante, Nidhal Carla Bouaynaya, **Ghulam Rasool**, and Lyudmila S. Mihaylova, “Towards Trustworthy Machine Learning – A Bayesian Framework,” DeepMath, San Diego, CA, November 17–18, 2022.
62. M. DelRocini, C. Angelini, and **Ghulam Rasool**, “Identification of Abnormalities in Head Computerized Tomography Scans,” The 2020 IEEE Signal Processing in Medicine and Biology Symposium, Philadelphia, PA, December 5, 2020.
<https://ieeexplore.ieee.org/abstract/document/9353610>.
63. Dimah Dera, **Ghulam Rasool**, and Nidhal Bouaynaya, “Robust Deep Learning Systems Integrated with Confidence Evaluation,” 6th Annual NJBDA Symposium, NJ City, NJ, April 2019.
64. Antonio Abbondandolo, Kiran Korah, Jason Wilkowski, Brandon Nugent, Nick Setaro, Erik Brewer, **Ghulam Rasool**, Nidhal Bouaynaya, and Emmanuel Zachariah, “LungPREDICT: Utilizing Artificial Intelligence to Differentiate Between Benign and Malignant Lung Cancer Tumors,” 2019 BMES Annual Meeting, Philadelphia, October 16–19, 2019.
65. Daniel Cahall, Nidhal Bouaynaya, **Ghulam Rasool**, Kiran Korah, Jason Wilkowski, Amanda Abruzzo, Erik Brewer, and Emmanuel Zachariah, “Classification of Lung Tumor Grade in CT Images Using Random Forest,” BMES/FDA Frontiers in Medical Devices Conference: The Role of Digital Evidence to Support Personalized Patient Healthcare, March 19–21, 2019, Washington, DC.
66. **Ghulam Rasool**, “Robust Deep Learning Systems Integrated with Confidence Evaluation,” DARPA AI Colloquium, Alexandria, Virginia, March 6–7, 2019.
67. Dimah Dera and **Ghulam Rasool**, “Robust Deep Learning Systems Integrated with Confidence Evaluation,” Rowan University Student Scholars Symposium, April 25–26, 2019.
68. Hikmat Khan, **Ghulam Rasool**, and Nidhal Bouaynaya, “Rotorcraft Flight Information Inference from Cockpit Videos Using Deep Learning,” Rowan University Student Scholars Symposium, April 25–26, 2019.
69. Christopher Angelini and **Ghulam Rasool**, “Landmark-less Head Pose Tracking for Pilots Using Recurrent Neural Networks: A Multi-Loss Approach,” Rowan University Student Scholars Symposium, April 25–26, 2019.
70. Shamoon Siddiqui and **Ghulam Rasool**, “Incremental Neural Architecture Search,” Rowan University Student Scholars Symposium, April 25–26, 2019.
71. **Ghulam Rasool**, Babak Afsharipour, Nina L. Suresh, and William Z. Rymer, “Alterations in Spatial Electromyogram Patterns of Hand Muscles in Hemiparetic Stroke Survivors,” Society for Neuroscience (SFN) Annual Meeting, San Diego, USA, November 12–16, 2016.
72. **Ghulam Rasool**, William Z. Rymer, Allison Wang, and Sabrina Lee, “Altered Rheological Properties of Passive Skeletal Muscles in Chronic Stroke,” Biomechanics and Neural Control of Movement (BANCOM), OH, USA, June 12–17, 2016.

73. **Ghulam Rasool**, William Z. Rymer, Allison Wang, and Sabrina Lee, “Changes in Viscoelastic Properties of Muscles in Chronic Stroke,” 2016 Congress of International Society of Electrophysiology and Kinesiology (ISEK), Chicago, USA, July 5–8, 2016.
74. **Ghulam Rasool**, Babak Afsharipour, Nina L. Suresh, and William Z. Rymer, “Analysis of Spatial Muscle Activation Patterns Post-Stroke,” 9th World Congress for NeuroRehabilitation (WCNR), Philadelphia, May 10–13, 2016.
75. Babak Afsharipour, Milap Sandhu, **Ghulam Rasool**, and William Z. Rymer, “Use of High-Density EMG Grid Recordings to Characterize the Level of Injury in Individuals Sustaining Cervical Spinal Cord Injury,” 2016 Congress of International Society of Electrophysiology and Kinesiology (ISEK), Chicago, USA, July 5–8, 2016.
76. **Ghulam Rasool**, Babak Afsharipour, Nina L. Suresh, Xiaogang Hu, and William Z. Rymer, “Altered Spatial Muscle Activation Patterns Reveal Possible Mechanisms of Motor Impairment in Stroke,” Society for Neuroscience (SFN) Annual Meeting, Chicago, USA, October 17–21, 2015.
77. **Ghulam Rasool** and Kamran Iqbal, “Muscle Synergies Based Task Discrimination for Myoelectric Control,” UALR Research and Creative Expo, UALR, 2014.
78. **Ghulam Rasool**, “Novel Methodology for Extraction of Control Information from the Myoelectric Signal Using the AR-GARCH Model,” UALR Research and Creative Expo, UALR, 2013.
79. Gregory S. Taylor, Yupo Chan, **Ghulam Rasool**, Richard Cronk, and James T. Moore, “A Three-Dimensional Bin-Packing Model: Exact Multicriteria Solution and Computational Complexity,” INFORMS Annual Meeting, Minneapolis, October 6–9, 2013.
80. **Ghulam Rasool**, Kamran Iqbal, and Gannon White, “Myoelectric Activity Detection During Sit-to-Stand Using Threshold Methods,” UALR Research and Creative Expo, UALR, 2012.

Book Chapter (Total Count: 1)

1. Asim Waqas, Dimah Dera, Ghulam Rasool, Nidhal Bouaynaya, and Hassan M. Fathallah-Shaykh, “Brain Tumor Segmentation and Surveillance with Deep Artificial Neural Networks,” in *Deep Learning for Biomedical Data Analysis*, edited by M. Elloumi, Springer, Cham, 2021. https://doi.org/10.1007/978-3-030-71676-9_13.

Non-Peer-Reviewed Publications (Total Count: 10)

1. Luis Felipe, Carlos Garcia, Issam El Naqa, Monique Shotande, Aakash Tripathi, Vivek Rudrapatna, **Ghulam Rasool**, Danielle Bitterman, Gilmer Valdes, “TheBlueScrubs-v1: A Comprehensive Curated Medical Dataset Derived from the Internet,” under review in Nature Scientific Data. Preprint available: <https://doi.org/10.48550/arXiv.2504.02874>.
2. Aakash Tripathi, Asim Waqas, Matthew B. Schabath, Yasin Yilmaz, **Ghulam Rasool**, “EAGLE: Efficient Alignment of Generalized Latent Embeddings for Multimodal Survival Prediction with Interpretable Attribution Analysis”. Preprint available: <https://arxiv.org/abs/2506.22446>.
3. Carlos Garcia-Fernandez, Luis Felipe, Monique Shotande, Muntasir Zitu, Aakash Tripathi, **Ghulam Rasool**, Issam El Naqa, Vivek Rudrapatna, Gilmer Valdes, “Trustworthy AI for Medicine: Continuous Hallucination Detection and Elimination with CHECK,” under review. Preprint available: <https://arxiv.org/abs/2506.11129>.
4. Asim Waqas, Aakash Tripathi, Paul Stewart, Mia Naeini, and **Ghulam Rasool**, “Embedding-based

Multimodal Learning on Pan-Squamous Cell Carcinomas for Improved Survival Outcomes,” preprint available: <https://doi.org/10.48550/arXiv.2406.08521>.

5. Linmin Pei, Granger Sutton, Michael Rutherford, Ulrike Wagner, Tracy Nolan, Kirk Smith, Phillip Farmer, Peter Gu, Ambar Rana, Kailing Chen, Thomas Ferleman, Brian Park, Ye Wu, Jordan Koujouharov, Gargi Singh, Jon Lemon, Tyler Willis, Milos Vukadinovic, Grant Duffy, Bryan He, David Ouyang, Marco Pereanez, Daniel Samber, Derek A. Smith, Christopher Cannistraci, Zahi Fayad, David S. Mendelson, Michele Bufano, Elmar Kotter, Hamideh Haghir, Rajesh Baidya, Stefan Dvoretzskii, Klaus H. Maier-Hein, Marco Nolden, Christopher Ablett, Silvia Siggillino, Sandeep Kaushik, Hongzhu Jiang, Sihan Xie, Zhiyu Wan, Alex Michie, Simon J Doran, Angeline Aurelia Waly, Felix A. Nathaniel Liang, Humam Arshad Mustagfirin, Michelle Grace Felicia, Kuo Po Chih, Rahul Krish, **Ghulam Rasool**, Nidhal Bouaynaya, Nikolas Koutsoubis, Kyle Naddeo, Kartik Pandit, Tony O’Sullivan, Raj Krish, Qinyan Pan, Scott Gustafson, Benjamin Kopchick, Laura Opsahl-Ong, Andrea Olvera-Morales, Jonathan Pinney, Kathryn Johnson, Theresa Do, Juergen Klenk, Maria Diaz, Arti Singh, Rong Chai, David A. Clunie, Fred Prior, Keyvan Farahani, “Medical Image De-Identification Benchmark Challenge”, Preprint available: <https://arxiv.org/abs/2507.23608>
6. Kyle Naddeo, Nikolas Koutsoubis, Rahul Krish, **Ghulam Rasool**, Nidhal Bouaynaya, Tony OSullivan, and Raj Krish, “DICOM De-Identification via Hybrid AI and Rule-Based Framework for Scalable, Uncertainty-Aware Redaction,” Preprint available: <https://arxiv.org/abs/2507.23736>.
7. Cahall, D. E., **Ghulam Rasool**, Bouaynaya, N. C., and Fathallah-Shaykh, H. M., “Dilated Inception U-Net (DIU-Net) for Brain Tumor Segmentation,” 2021. Available online: <https://arxiv.org/abs/2108.06772>.
8. Nesrine Amor, **Ghulam Rasool**, and Nidhal C. Bouaynaya, “Constrained State Estimation – A Review,” 2018. Available online: <https://arxiv.org/abs/1807.03463>.
9. Nikolas Koutsoubis, Yasin Yilmaz, Matthew Schabath, and **Ghulam Rasool**, “Privacy-Preserving Federated Learning in Medical Imaging with Uncertainty Estimation,” 2024. Available online: <https://doi.org/10.48550/arXiv.2406.12815>.
10. Jacob Epifano, Ravi Ramachandran, and **Ghulam Rasool**, “Deployment of a Robust and Explainable Mortality Prediction Model: The COVID-19 Pandemic and Beyond,” 2023. Preprint available: <https://doi.org/10.48550/arXiv.2311.17133>.

Patents (Total Count: 2)

1. Hassan Fathallah-Shaykh, Nidhal Bouaynaya, **Ghulam Rasool**, and Dimah Dera, “Method for Uncertainty Estimation in Deep Neural Networks,” International Patent Application No: PCT/US2020/053441, filed on September 30, 2020. Link: <https://patents.google.com/patent/US20220366223A1/en>.
2. Gilmer Valdes, Carlos Garcia Fernandez, Luis Felipe, Aakash G. Tripathi, **Ghulam Rasool**, Issam M. El Naqa, and Monique O. Shotande, “Large Language Model Certainty Determination,” patent pending.

Invited Talks, Seminars and Panel Presentations (Total Count: 30)

1. Invited Presenter and Panelist, Cancer Informatics for Cancer Centers (CI4CC) 2025, Session 6 “The Digital Bridge to Precision Health: Wearables, Real-World Data, and AI”, title of the talk “AI-Driven Early Detection of Cancer Cachexia in Pancreatic Cancer”, Miami, FL, November 7–9, 2025.

2. Invited Speaker, Health Choice Network (HCN) AI Forum, “Research Engagement Forum – Chatbots and Patient Engagement”, Online, September 24, 2025.
3. Invited Speaker, AlicanceChicago and Health Choice Network (HCN), Research Engagement Forum, “Making Chatbots Safe for Cancer Survivors: An update on data collection and preliminary findings”, Online, November 3, 2025.
4. Invited Panelist, Florida Cancer Innovation: Nourish to Flourish Against Cancer A Florida Cancer Connect Collaborative World Cancer Research Day Roundtable, “AI-Driven Early Detection of Cachexia in Pancreatic Cancer and Feasibility of Diet and Exercise Interventions”, University of South Florida Morsani College of Medicine, Tampa, FL, September 24, 2025.
5. Invited Panelist, “AI Driven Breakthroughs and the Future of Pediatric Cancer Care, 5th Annual Live Like Bella Pediatric Cancer Research Symposium, Miami, FL, September 12, 2025.
6. Invited Panelist, “General Session – From Principles to Practice: Leading AI Integration with Purpose in Community Health,” Health Choice Network (HCN), 31st Annual Board Education Conference, Marco Island, FL, USA, June 19–21, 2025.
7. Webinar Co-Presenter, “Immersive Visualization and Interventions – A Multicenter Experience,” Society for Imaging Informatics in Medicine (SIIM) Enterprise Imaging Webinar, Virtual, April 29, 2025.
8. Invited Speaker, “From Reports to Reasoning: Leveraging Large Language Models for Pathology and Radiology Workflows,” Pistoia Alliance NLP Group Monthly Meeting, Virtual, May 26, 2025.
9. Invited Speaker, “From Reports to Reasoning: Leveraging Large Language Models for Pathology and Radiology Workflows,” Nepal Mathematical Society (NMS) and Association of Nepalese Mathematicians in America (ANMA) Talk Series, Virtual, May 18, 2025.
10. Invited Panelist, “Digital Cytology/AI,” JASC Panel Session, 72nd Annual Scientific Meeting, Digital Pathology Association / American Society of Cytopathology (ASC), Orlando, FL, USA, November 6–10, 2024.
11. Presenter, “Multimodal Foundation Models for Predicting Overall Survival in Pan-Cancer Settings,” Dr. Robert Gillies Machine Learning Workshop in Cancer, Moffitt Cancer Center, Tampa, FL, USA, November 14–15, 2024.
12. Invited Panelist, “Cancer AI,” University of Florida Data and Artificial Intelligence Symposium (DAISY), One Ocean Resort and Spa, Jacksonville Beach, FL, USA, September 9–10, 2024.
13. Invited Speaker, “Large Language Models,” Neuro-Oncology Retreat, Department of Neuro-Oncology, Moffitt Cancer Center, Tampa, FL, USA, 2024.
14. Invited Speaker, “Large Language Models and Their Potential Role in Future Healthcare Systems,” Advanced Digital Literacy Center, University of Tampa, Tampa, FL, USA, February 2024.
15. Invited Speaker, “Machine Learning and CT Image Analyses for the Segmentation of Skeletal Muscles and Fat,” ColoCare Investigator Meeting, Heidelberg University, Virtual, November 2, 2023.
16. Invited Speaker, “Artificial Intelligence for Cancer Care,” University of Tampa, Tampa, FL, USA, November 2, 2023.
17. Invited Speaker, “Towards Building Trustworthy Machine Learning Models,” University of Texas Southwestern Medical Center, Virtual, February 2023.
18. Invited Speaker, “Trustworthy Machine Learning,” Department of Health Outcomes and Behavior (HOB) Meeting, Moffitt Cancer Center, Tampa, FL, USA, December 2022.

19. Invited Speaker, “Brain Tumor Surveillance with Trustworthy Machine Learning,” ML Memorial Workshop for Dr. Gillies, Clearwater, FL, USA, November 2022.
20. Invited Speaker, “Trustworthy AI for Brain Tumor Segmentation,” Neuro-Oncology Translational Research Interest Group (NOTRIG) Meeting, Moffitt Cancer Center, Tampa, FL, USA, June 2022.
21. Invited Speaker, “Trustworthy Machine Learning,” 6th Annual SJ Neurovascular and Stroke Symposium, The Mansion on Main Street, Voorhees, NJ, USA, May 25, 2022.
22. Invited Speaker, “Towards Lifelong Learning with Self-Aware Neural Networks,” Big Data Working Group (BigDAWG), Federal Aviation Administration (FAA), Virtual, April 2020.
23. Invited Speaker, “AI in Imaging & Radiation Oncology,” AAPM Spring Symposium, Philadelphia, PA, USA, May 2019.
24. Invited Speaker, “Deep Learning: Is It the Answer to AI?,” Rowan Faculty Research Day, Rowan University, Glassboro, NJ, USA, March 2019.
25. Invited Speaker, “Bringing Deep Learning into Neuro-Oncology – Tumor Delineation and Surveillance,” MD Anderson Cancer Center, Camden, NJ, USA, January 2019.
26. Invited Speaker, “Machine Learning and Data Analytics: Moving AI into the Future,” Lockheed Martin Internal Conference, Virtual (hosted from Syracuse, NY), October 2018.
27. Invited Speaker, “Enhancing Myoelectric Control Using Muscle Synergies,” Rehabilitation Institute of Chicago, Chicago, IL, USA, January 2014.
28. Invited Speaker, “Myoelectric Controlled Powered Prostheses – Recent Results,” IEEE Little Rock Chapter Meeting, Little Rock, AR, USA, 2013.
29. Guest Speaker, “Application of the EMG Signal in Engineering,” HHPS-7323 Advanced Biomechanics Course, Department of Health, Human Performance and Sports Management, University of Arkansas at Little Rock, Little Rock, AR, USA, 2013.
30. Invited Speaker, “Myoelectric Controlled Powered Prostheses – Present Trends and Future Prospects,” IEEE Little Rock Chapter Meeting, Little Rock, AR, USA, October 2012.

Service

Institutional Service - Moffitt Cancer Center

- 2024–Present: Member, Moffitt Conflict Committee.
- 2023–Present: Member, Moffitt HPC Committee.
- 2025–Present: Member, Research Domain Council.
- 2025–Present: Co-Chair, Clinical AI Committee.
- 2025–Present: ACD-PQS Search Committee.
- 2025–Present: Chair, Departmental Faculty Search Committee.
- 2024–Present: Member of Organizing Committee, Gillies Machine Learning Workshop in Cancer.
- 2024–Present: Reviewer, Junior Scientist Research Partnership (JSRP) Review Panel.
- 2022–Present: Member, Moffitt Research IT Council.

- 2023–2024: Member, ML Vice Chair Search Committee.
- 2023–Present: Faculty representative for the Remission AI Cluster.

Institutional Service - Rowan University

- 2019–2020: Member, Faculty Search Committee.
- 2021–2022: Member, Curriculum Committee.
- 2021: Member, Anti-Racist Pedagogy and Practices Task Force.

Professional Service

- 2024–Present: Member, University of Tampa CS Industry Advisory Board.
- 2025–Present: Member, AI Curriculum Committee, Digital Pathology Association (DPA).
- 2024–Present: Member, Society for Imaging Informatics in Medicine (SIIM) Machine Learning Research & Tools Committee
- 2024–2025: Member, Convergent Science Section, Bioinformatics, Computational Biology, Systems Biology, and Convergent Science Subcommittee, Program Committee for the AACR Annual Meeting 2025.
- 2024: Associate Editor, *British Journal of Radiology (BJR)*.
- 2024: Reviewer, NIH Panel ZRG1 HSS-N (90).
- 2024: Reviewer, Swiss National Science Foundation.
- 2024: Reviewer, Czech Science Foundation.
- 2023–2024: Guest reviewer, *Nature Methods*, *Cell*.
- 2022: Guest reviewer, *Journal of Alzheimer's Disease*.
- 2022: Guest reviewer, *IEEE Transactions on Biomedical Engineering*.
- 2020–2021: Reviewer, National Science Foundation (NSF).
- 2019–2021: NSF XSEDE Campus Champion.
- 2019–Present: NVIDIA Ambassador for Rowan University.
- 2020–2021: Special Session and Program Chair, International Joint Conference on Neural Networks (IJCNN).
- 2020–2021: Guest reviewer, *Neural Networks*.
- 2020: Program Chair, Practice and Experience in Advanced Research Computing.
- 2020: Guest reviewer, *IEEE Transactions on Neural Systems and Rehabilitation Engineering*.
- 2018: Guest reviewer, International Conference on Control, Engineering & IT (CEIT).
- 2018: Guest reviewer, *IEEE Access*.
- 2016: Guest reviewer, *Journal of NeuroEngineering and Rehabilitation*.
- 2016: Guest reviewer, *IEEE Transactions on Biomedical Engineering*.

- 2016: Guest reviewer, *IEEE Engineering in Medicine and Biology Conference*.
- 2016: Guest reviewer, *PLOS One*.

Professional Association Memberships

- 2014–Present: Member, Institute of Electrical and Electronics Engineers (IEEE).
- 2022–Present: Member, American Association for Cancer Research (AACR).
- 2022–Present: Member, Digital Pathology Association (DPA).
- 2021–Present: Member, Society for Imaging Informatics in Medicine (SIIM).
- 2024–Present: Affiliate Member, Society for Advanced Body Imaging (SABI).
- 2024–Present: Affiliate Member, United States and Canadian Academy of Pathologists (USCAP).

Teaching Experience

Short Courses

- Spring 2027–2029, “Essentials of Generative Artificial Intelligence for Residents, Fellows, Pathologists, and other Laboratory Professionals”, accepted by the United States and Canadian Academy of Pathology (USCAP) for launch in the 2027 annual meeting, Vancouver, BC, Canada.

University Courses

- Spring 2022: Course Instructor, Systems and Control – I (ECE 09321-2); Engineering Clinics (ENGR 01303/01403), two sections: (1) Gesture Control for Mixed Reality and (2) VR Modeling for Landfills.
- Fall 2021: Course Instructor, Reinforcement Learning (ECE 09458-1 and ECE 09504-3); Engineering Clinics (ENGR 01303/01403), Prosthetic arm – electrical, mechanical, and machine learning design.
- Spring 2021: Course Instructor, Digital Signal Processing (ECE 09351-3); Engineering Clinics (ENGR 01303/01403), two sections: (1) Prosthetic arm design and (2) Deep learning for the real world – natural language processing (NLP) applications.
- Fall 2020: Course Instructor, Deep Learning (ECE 09595 and ECE 09494); Engineering Clinics (ENGR 01303/01403), two sections: (1) Prosthetic arm design and (2) Deep learning for the real world – adversarial attacks and defenses.
- Spring 2020: Course Instructor, Digital Signal Processing (ECE 09351-3); Engineering Clinics (ENGR 01303/01403), two sections: (1) Prosthetic arm design and (2) Deep learning for the real world.
- Fall 2019: Course Instructor, Reinforcement Learning (ECE 09595 and ECE 09494); Engineering Clinics (ENGR 01303/01403), two sections: (1) Prosthetic arm design and (2) Deep learning for the real world.
- Spring 2019: Course Instructor, Deep Learning (ECE 09595 and ECE 09494); Digital Signal Processing (ECE 09351); Systems and Control – I (ECE 09321); Engineering Clinics (ENGR 01303/01403), two sections: (1) Prosthetic arm – electrical design and (2) Prosthetic arm – mechanical design.
- Fall 2018: Course Instructor, Introduction to Digital Systems (ECE 09241), three sections; Signals and Systems (ECE 09341); Engineering Clinics (ENGR 01303/01403), three sections: (1) Prosthetic arm – electrical design and (2) Prosthetic arm – mechanical design.

- Spring 2018: Course Instructor, Deep Learning (ECE 09595 and ECE 09494); Signals and Systems (ECE 09341); Engineering Clinics (ENGR 01303/01403), three sections: (1) Prosthetic arm – electrical design, (2) Prosthetic arm – mechanical design, and (3) Learning to Deep Learn.
- Fall 2017: Course Instructor (remote co-instructor with Dr. Bouaynaya), Engineering Clinics (ENGR 01303/01403), two sections: (1) Prosthetic arm – electrical design and (2) Prosthetic arm – mechanical design.

National and Local Workshops

- July 7–8, 2025: Instructor, “Scalable Multimodal AI in Oncology Using HONeYBEE: From Embeddings to Clinical Impact”, Mayo AI Summit 2025, Rochester, MN.
- February 22, 2024: Instructor, “Building Transformer-based Natural Language Processing,” NVIDIA Deep Learning Institute, Public Student Workshop, North America, Virtual.
- November 3, 2023: Instructor, “Building Transformer-based Natural Language Processing,” NVIDIA Deep Learning Institute, Public Student Workshop, North America, Virtual.
- June 29, 2022: Instructor, “Building Transformer-based Natural Language Processing,” NVIDIA Deep Learning Institute, Public Student Workshop, North America, Virtual.
- August 9–11, 2021: Instructor, “Fundamentals of Deep Learning,” Workshop at the 64th IEEE International Midwest Symposium on Circuits and Systems (MWSCAS 2021), Virtual.
<https://www.mwscas2021.org/workshop-and-tutorials>.
- February 22, 2020: Instructor, “Fundamentals of Deep Learning for Computer Vision,” NVIDIA Sponsored Workshop, Rutgers Business School, Rutgers University, NJ, USA.
- November 30, 2018: Instructor, “Fundamentals of Deep Learning for Computer Vision,” NVIDIA Sponsored Workshop, Rowan University, Glassboro, NJ, USA.
- November 16, 2018: Instructor, “Fundamentals of Deep Learning for Computer Vision,” NVIDIA Sponsored Workshop, Rowan University, Glassboro, NJ, USA.
- October 11, 2018: Instructor, “Machine Learning and Data Analytics – I: Data, Big Data and Analytics,” Lunch & Learn talk at Lockheed Martin, Virtual.
- October 25, 2018: Instructor, “Machine Learning and Data Analytics – II: Data, Big Data and Analytics,” Lunch & Learn talk at Lockheed Martin, Virtual.
- November 2018: Instructor, “Machine Learning and Data Analytics – III: Data, Big Data and Analytics,” Lunch & Learn talk at Lockheed Martin, Virtual.

Mentoring and Training Experience

High School Students

- 2022–2024: Kavya Venkat, Old Bridge High School, NJ.
- 2020–2021: Mentored multiple high school students, including Kavya Venkat, Bhavini Pandey, Lucas Eberhardt, and Neal Patel, from different high schools in NJ.

Undergraduate Students

- 2025: Kavya Venkat.

- 2024: Kavya Venkat, Isabel Ung, Narmeen Sabah Siddiqui, and Ameer Abdul Rafay, Department of Machine Learning, Moffitt Cancer Center.
- 2023: Sabrina Khan, Isabel Ung, Muhammad Ali Afridi, Department of Machine Learning, Moffitt Cancer Center.
- 2018–2022: Several undergraduate students, Department of Electrical and Computer Engineering, Rowan University.
- 2016: Hilda MA Zhiyao, Rehabilitation Institute of Chicago and Northwestern University, Chicago, IL.

Graduate Students – Under Training

1. 2025–Continued: Major Professor, Ph.D. Graduate Student Supervisory Committee for Ms. Hanieh Ajami, Department of Computer Science and Engineering, University of South Florida.
2. 2023–Continued: Major Professor, Ph.D. Graduate Student Supervisory Committee for Mr. Nikolas Koutsoubis, Electrical Engineering, University of South Florida.

Graduate Students – Graduated

1. 2022–2025: Major Professor, Ph.D. Graduate Student Supervisory Committee for Mr. Aakash Tripathi, Electrical Engineering, University of South Florida.
2. 2020–2025: Committee Member, Ph.D. Graduate Student Supervisory Committee for Mr. Ian Nielsen, Electrical and Computer Engineering, Rowan University.
3. 2020–2025: Major Professor, Ph.D. Graduate Student Supervisory Committee for Ms. Sabeen Ahmed, Electrical Engineering, University of South Florida.
4. 2020–2024: Major Professor, Ph.D. Graduate Student Supervisory Committee for Mr. Asim Waqas, Electrical Engineering, University of South Florida.
5. 2021–2023: Major Professor, MS Supervisory Committee for Mr. Yassine Barhoumi, Electrical and Computer Engineering, Rowan University.
6. 2021–2022: Major Professor, MS Supervisory Committee for Mr. Abdullah Nasir, Electrical and Computer Engineering, Rowan University.
7. 2023–2024: Mentor, Mr. Ashwin Mukund, Graduate Trainee, Moffitt Cancer Center; currently at Johns Hopkins University.
8. 2019–2023: Member, Ph.D. Graduate Student Supervisory Committee for Ms. Giuseppina Carannante, Electrical and Computer Engineering, Rowan University.
9. 2019–2023: Co-Advisor (with Dr. Ravi Ramachandran), Ph.D. Graduate Student Supervisory Committee for Mr. Jacob Epifano, Electrical and Computer Engineering, Rowan University.
10. 2023: Member, Ph.D. Graduate Student Supervisory Committee for Mr. Glenn Dawson, Electrical and Computer Engineering, Rowan University.
11. 2023: Member, Advisory Committee, Ph.D. Graduate Student Supervisory Committee for Mr. Muhammad Umer, Electrical and Computer Engineering, Rowan University.
12. 2022: Member, Advisory Committee, Ph.D. Graduate Student Supervisory Committee for Mr. Usman Ghumman, National University of Science and Technology (NUST), Islamabad, Pakistan.

13. 2018–2024: Member, Ph.D. Graduate Student Supervisory Committee for Mr. Hikmat Khan, Electrical and Computer Engineering, Rowan University.
14. 2019–2024: Member, Ph.D. Graduate Student Supervisory Committee for Mr. Christopher Angelini, Electrical and Computer Engineering, Rowan University.
15. 2020–2021: Member, MS Supervisory Committee for Mr. Mahmut Gemici, Electrical and Computer Engineering, Rowan University.
16. 2019–2021: Member, MS Supervisory Committee for Mr. David Specht, Electrical and Computer Engineering, Rowan University.
17. 2018–2020: Member, MS Supervisory Committee for Mr. Eric Feuerstein, Electrical and Computer Engineering, Rowan University.
18. 2016–2018: Member, MS Supervisory Committee for Mr. Oliver Palumbo, Electrical and Computer Engineering, Rowan University.
19. 2015–2016: Member, MS Supervisory Committee for Mr. Hamdi Albusnashee, University of Arkansas at Little Rock, Little Rock, AR.

Post-Doctoral Trainees

1. 2024–Continued: Dr. Asim Waqas, Department of Cancer Epidemiology, Moffitt Cancer Center.
2. 2023–Continued: Dr. Iryna Hartsock, Department of Machine Learning, Moffitt Cancer Center.
3. 2020–2021: Dr. Dimah Dera, Department of Electrical and Computer Engineering, Rowan University.

Other Education

- January 2018 – March 2018: NSF I-Corps Entrepreneurial Lead.
- July 2018: Google Cloud Platform Fundamentals: Core Infrastructure (Online).
- July 2018: How Google Does Machine Learning (Online).
- 2015: COMSOL Workshop.
- 2012: Application of New Technologies in Rehabilitation.